OPEN AGENT HACKATHON 2026

Ledger Lens

An AI finance team for freelancers.

It finds where the money leaks, proves every finding with data, and recommends three moves a human approves or rejects.

- Track 03 — Reasoning Architecture
- Zetaris · Meterless · NVIDIA Nemotron

01	**Leak Hunter** 6 SQL checks + ranking	Nemotron 3.5 Lightning
02	**Explainer** cited causes	Nemotron 3 Ultra Markovian
03	**Forecaster** 6-month cash + what-ifs	deterministic TypeScript
04	**CFO** 3 ranked moves	Nemotron 3 Ultra H-MEM

THE PROBLEM

Sara has nine clients and no CFO

SC

Sara Cohen

Freelance brand & UI designer · Casablanca

9 clients | 54 invoices | 6 months | EUR

Five tables of history: clients, invoices, time logs, subscriptions, and bank transactions. The answers are there, spread across all five.

> "My accounting tool tells me what happened. It never tells me what to do next."

01 · UNBILLED HOURS

€1,320/mo

22 Atlas Co hours never invoiced

02 · QUIET PRICE RISE

€14/mo

Adobe +20.9%: €66.99 → €80.99

03 · UNUSED TOOL

€29/mo

Webflow unopened for 94 days

04 · LATE PAYER

€4,800

Two Nordwind invoices, 33–47 days late

05 · UNDERPRICED SERVICE

€540/mo

Kestrel at €42/h vs €60/h market

06 · DUPLICATE CHARGE

€15/mo

Figma charged twice in August

€1,918
/MO
RECURRING LEAKS

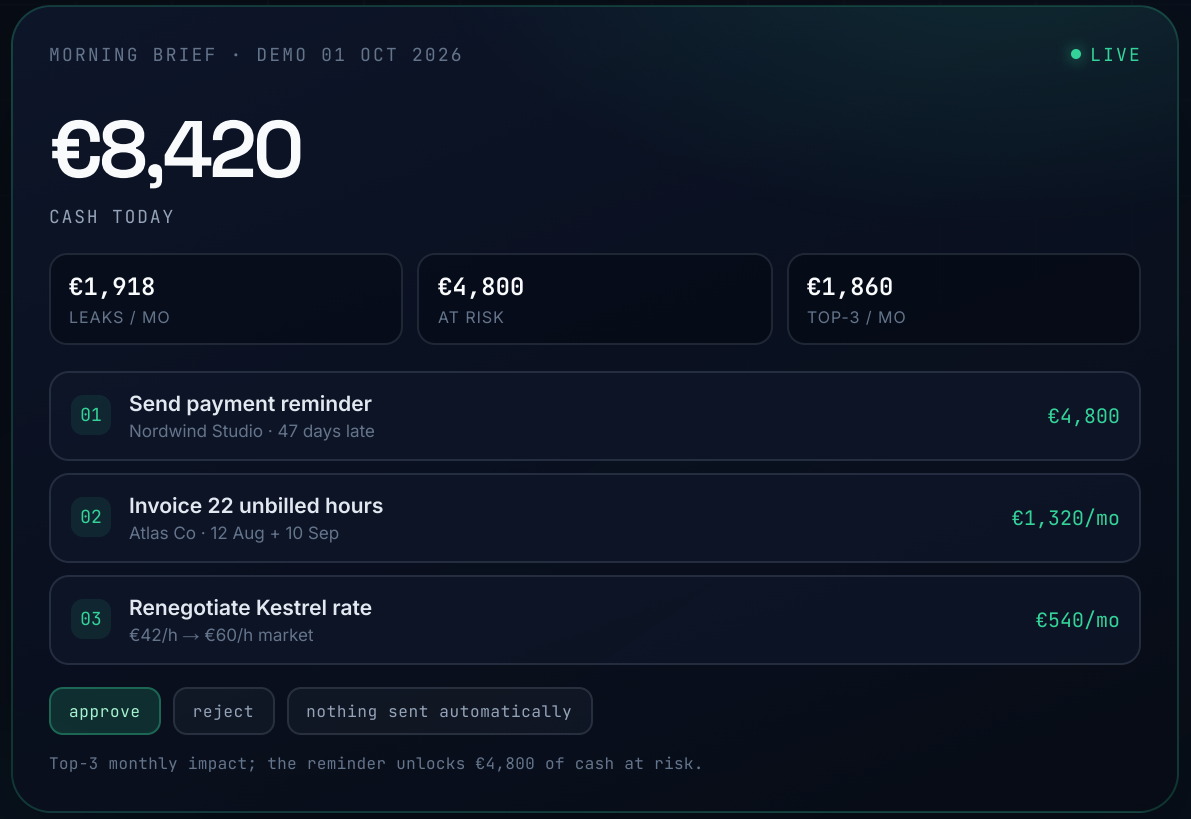
€4,800
CASH AT RISK

6
PLANTED LEAKS · 0 FALSE ALARMS

SOURCE: DATA/ANSWER_KEY.JSON · DATA/OUT/SUMMARY.JSON

THE SOLUTION

A morning brief, not another dashboard



01 Six findings, each with proof
Every finding stores the Zetaris SQL and the rows behind it.

02 Three ranked moves
Each move has a EUR impact, an action type, and a reason.

03 Human keeps the decision
Sara approves or rejects; nothing is sent automatically.

04 Memory changes the next run
H-MEM remembers the decision and re-ranks the next brief.

05 Numbers never come from the model
Every amount is read from a Zetaris view, not generated.

BRIEF → LEAK → EVIDENCE → WHY → WHAT-IF → DRAFT → REJECT → RERUN + MEMORY

SOURCE: README.MD · DOCS/CONTRACTS.MD §3

HOW THE AGENTS WORK

Four agents, one evidence trail

01 FIND

Leak Hunter

Six fixed SQL checks find the leaks; the fast model ranks and describes them.

`Nemotron 3.5 Lightning`

→ findings f001–f006

02 EXPLAIN

Explainer

Markovian reasoning on the deep model; every cause cites a finding.

`Nemotron 3 Ultra · Markovian`

→ cited explanation

03 FORECAST

Forecaster

Deterministic cash math in code, not a model. Six-month projection and what-ifs.

`TypeScript · no model`

→ 6-month cash + what-ifs

04 DECIDE

CFO

Deep model ranks three moves; H-MEM changes the ranking from past decisions.

`Nemotron 3 Ultra · H-MEM`

→ 3 moves + memory note

Evidence + trace

6 findings · 3 moves · one line per step and model call

`logs/<run_id>.jsonl`

CRITIC LOOP

An answer that cites an amount not in the evidence goes back to Nemotron once, with the allowed amounts; if it is still wrong, the deterministic fallback is used. Seen live as a `critique` line in the trace.

THE DATA LAYER

Agents never touch raw tables

SOURCE 01 · POSTGRES

ledger_lens_pg

5 tables · clients, invoices, time logs, subscriptions, bank transactions

SOURCE 02 · PUBLIC REST

fx_rest

EUR snapshot · CC0 1.0 licence

→

FEDERATED QUERY LAYER

Zetaris

container ledger_lens · SELECT-only

4 views 1 read path querySql

→

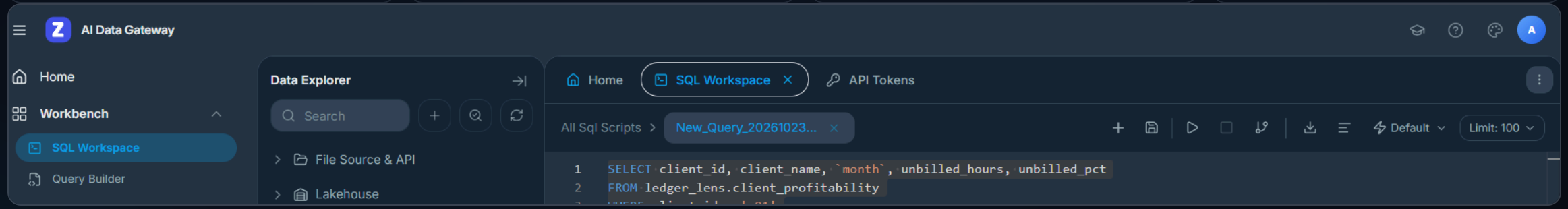
- **client_profitability** — unbilled hours, rates, margin
- **cash_position** — cash today + runway
- **subscription_usage** — price changes, logins, duplicates
- **ar_aging** — overdue invoices + aging buckets

Read path
querySql is the only client

Evidence
SQL + rows stored on every finding

Setup
./zetaris/setup_zetaris.sh · dry-run + verify-only

Modes
ZETARIS_MOCK=1 fixtures · 0 live



THE REASONING STACK

Meterless engines + NVIDIA Nemotron

Fast lane

rank + describe

Leak Hunter ranks and describes the six findings.

Nemotron 3.5 Lightning

Output: ranked findings f001–f006

Deep lane

explain + decide

Explainer and CFO produce cited causes and three ranked moves.

Nemotron 3 Ultra

Output: explanation + three ranked moves

Build: TypeScript + Fastify · React + Vite + Tailwind · Postgres 16 · Docker · Zetaris · Meterless Nightshift

REASONING LOOP — EXPLAINER

Markovian

Bounded step-by-step investigation; evidence is carried forward.

inspect → cite → check → answer

MEMORY — CFO

H-MEM

Remembers approve/reject decisions and changes the next ranking.

decision → memory note → re-rank

SHARED STATE — AGENTS

World Model

Each run ingests the findings and the full view rows; it powers the Client 360 drawer.

79 entities · clients, subscriptions, invoices

GROUNDING RULE A model may only cite amounts that appear in the finding or its evidence rows. If Nightshift is offline, deterministic fallbacks keep the same numbers. Full trace: `logs/<run_id>.jsonl`.

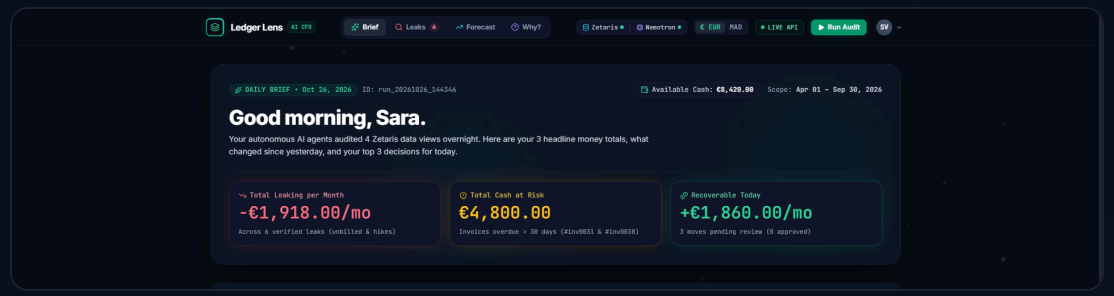
TRACE run_20261026_181940 → leak_hunter → explainer → forecaster → cfo

6 FINDINGS · 3 MOVES · LIVE ZETARIS + NEMOTRON

SOURCE: DOCS/ARCHITECTURE.MD · METADATA.JSON · DOCS/ENGINE-TESTS.MD

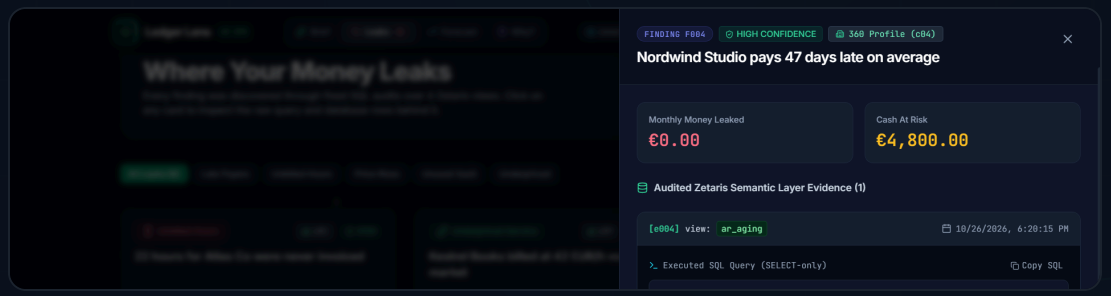
THE PRODUCT

Four screens that tell the story



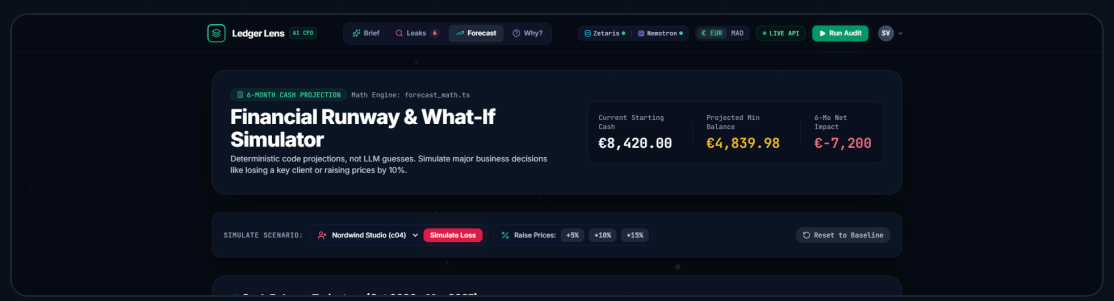
Brief

01 · MONEY HEADLINE + MOVES



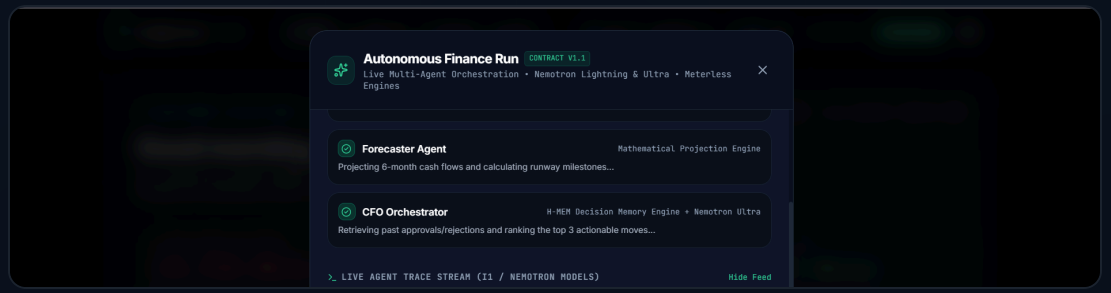
Leaks + evidence

02 · SQL + ROWS



Forecast

03 · SIX-MONTH + WHAT-IFS



Why? / timeline

04 · CITED CAUSES + TRACE

LIVE RUN · 26 OCT 2026 · DRAFTS OPEN IN A MODAL MARKED “NEVER SENT AUTOMATICALLY”

PROOF

The numbers are checkable

6/6	0	1,256	165
PLANTED LEAKS FOUND	FALSE ALARMS	DATA SANITY CHECKS	BACKEND TESTS

€1,918	€4,800	€8,420
/mo Recurring leaks	Cash at risk	Cash today

```
$ python evals/check_findings.py --findings examples/findings.json
6/6 found, 0 false alarms
$ python evals/stability.py --runs 5
PASS: 5/5 live runs · 6/6 each · 0 llm_failed
$ cd backend && npm test
165 passed (165)
```

Dataset: 9 clients · 10 subscriptions · 54 invoices · 132 time logs · 168 transactions · 942 billable hours

WHAT IS UNIQUE

Not a chatbot over a spreadsheet

01

Federated data

Postgres + a public FX source behind one Zetaris layer.

4 views · 1 read path

02

Grounded reasoning

SQL finds facts; models rank, explain, and choose wording; code does the forecast.

every amount traced to a view row

03

Visible engines

World Model, H-MEM, and Markovian are tested integrations — not prompt labels.

10 tests per engine

04

Human in the loop

Every move waits for approval, and the decision changes the next run.

decision → memory → next run

05

Honest fallback

Without Nightshift, the same six leaks and numbers still work.

6/6 leaks without an API key

ZETARIS VIEWS → EVIDENCE → GROUNDED MODEL OUTPUT → DETERMINISTIC MATH → HUMAN DECISION → MEMORY CHANGES NEXT RUN

Most finance demos stop at “the model read the data and wrote a summary.” Ledger Lens stops at a decision.

FUTURE WORK

From a synthetic demo to a real finance team

NOW · IN THE DEMO

The loop works

- Synthetic six-month dataset
- Four Zetaris views + public FX source
- Six findings, three ranked moves
- Approve/reject + H-MEM memory

NEXT · REAL DATA

Connect the sources

- Real bank data through Open Banking
- Invoicing and time-tracker connectors
- Recurring charges detected from bank data
- Usage from Google/Microsoft or vendor APIs

LATER · PRODUCT

Make it a finance team

- One database per user
- Hosted AI instead of a desktop runtime
- Receipts and partial payments

The architecture is already shaped for them: a new source becomes a new Zetaris source and view, not a rewrite of the agents.